Zero Sum Subarray

AlgoExpert · Medium · Arrays

The problem

- Does any non-empty contiguous subarray sum to zero?
- A repeated prefix sum proves it.

Intuition

- If two prefix sums are equal, the elements between them sum to zero.

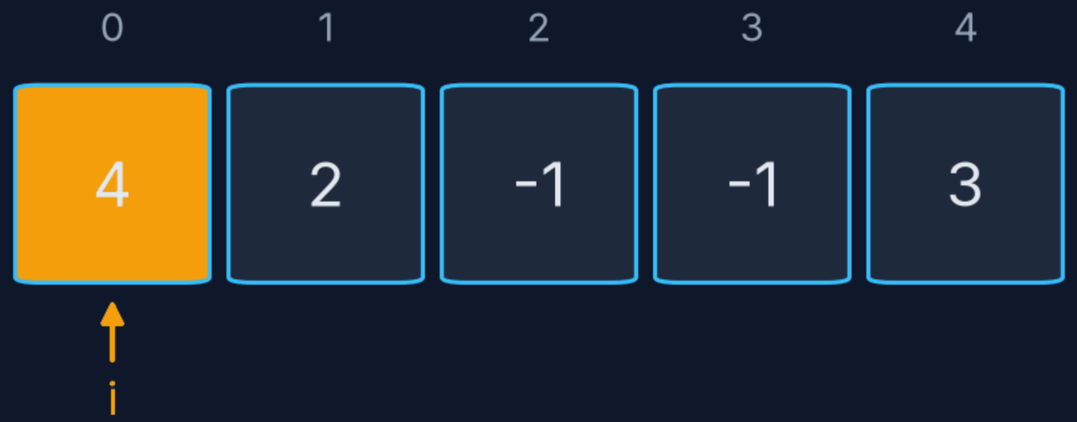
Brute force vs optimal

- Brute force: all subarrays → $O(n^2)$ (or $O(n^3)$ naive sums)
- Prefix sums + hash set → $O(n)$
- Repeated prefix ⇒ zero-sum in between

Approach

- 1. Keep a set of prefix sums starting with zero
- 2. Add each element to the running sum
- 3. If the sum was seen before, return true

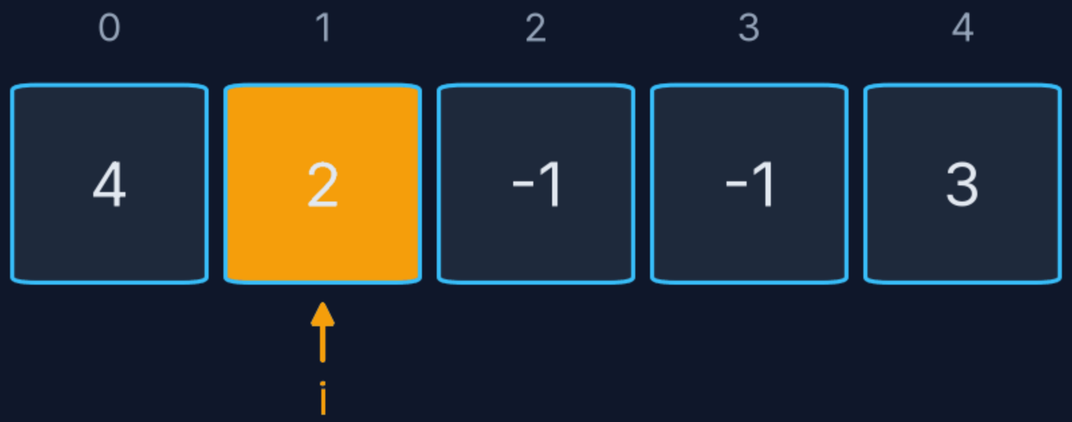
Worked example




```
prefix sum = 4
seen = {0}
```

Prefix sum 4 is new

Worked example




```
prefix sum = 6
seen = {0, 4}
```

Prefix sum 6 is new

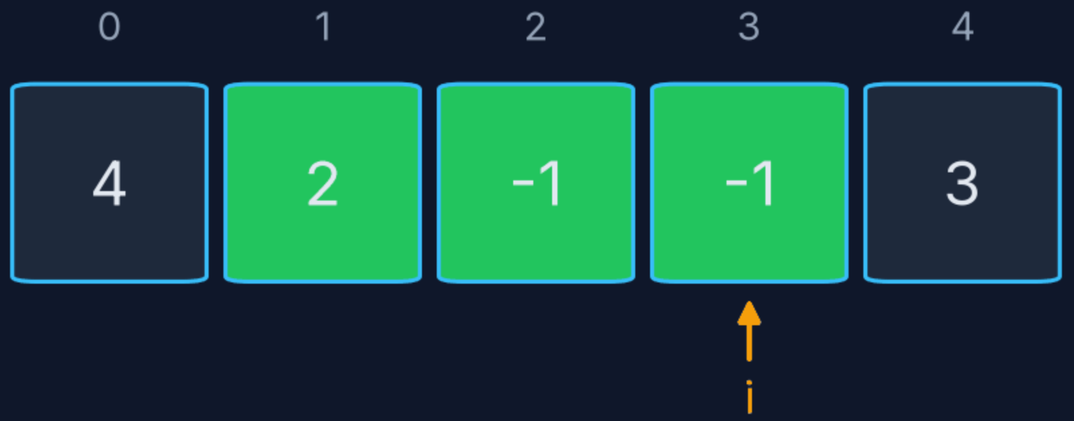
Worked example




```
prefix sum = 5
seen = {0, 4, 6}
```

Prefix sum 5 is new

Worked example




```
prefix sum = 4
seen = {0, 4, 6, 5}  ✓ repeat!
```

Prefix 4 seen before → zero-sum subarray

Java solution

```java
public class ZeroSumSubarray {
    public static boolean zeroSumSubarray(int[] nums) {
        Set<Long> seen = new HashSet<>();
        seen.add(0L);
        long sum = 0;
        for (int n : nums) { sum += n; if (!seen.add(sum)) return true; }
        return false;
    }
}
```

Complexity

- Time: O(n)
- Space: O(n)

Edge cases & pitfalls

- Seed the set with 0 (subarray starting at index 0)
- A single 0 element is a zero-sum subarray
- Use long for sums

Interview tips & follow-ups

- Follow-up: count zero-sum subarrays (map of counts)
- Follow-up: subarray sum = k

Recap

- If two prefix sums are equal, the elements between them sum to zero.
- Complexity: O(n)